A Lidar Benchmark for Desert Rover Traversability: Slip Face Detection, Hazard Layers and Route Planning from Coarse Elevation Models on Active Sand Dunes

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***Abstract*—For most deserts the only free elevation model a small wheeled rover can plan on has a posting of about 30 m, while the main hazard on active dunes, the slip face, is often only metres long. We built a benchmark on six North American dune fields with one metre airborne lidar as truth. Ten families of hazard layers, from slope and interpolation to boosted trees, a convolutional network, learned super resolution and imagery fusion, were scored against a lidar face map on 10 and 30 m block means (four of them from 2 to 60 m) and three real elevation products, and used by a slip aware planner at operating points chosen on held out sites. At 10 m the best layer raises mission success from 17 % to 64 %; at 30 m no layer we planned with adds more than 5 points to 8 %, because most faces are shorter than one cell, while a face map from fine data would reach 81 %. Onboard sensing that sees every face within 20 m keeps the rover off faces whatever the prior, but in a post hoc check with line of sight the rover succeeded on 30 % of the pairs with the slip model prior and 74 % with a face map prior. Code, results and the prediction log are public.**

***Index Terms*—field robotics, traversability, airborne lidar, digital elevation models, sand dunes, slip faces, hazard mapping, path planning, benchmark**

I. Introduction

Small wheeled ground robots are proposed for survey, inspection and search in sand deserts, including the dune seas of the Arabian Peninsula. Before such a robot drives, its route is planned on a prior map, and for most deserts the only free digital elevation models (DEMs) are global products with a posting of about 30 m [1], [2], [3]. The dominant hazard on active dunes is the slip face, the lee side of a dune held near the angle of repose of dry sand, about 30 to 34 degrees [4], where wheel slip rises until a small vehicle can no longer climb [5], [6]. Many slip faces are shorter than 30 m (Section IV-B), so a 30 m cell can hold one while its mean slope is gentle (Fig. 1).

That a coarse grid understates slope is long known [7]. What is not known is how much this matters for a planner, which hazard layer a practitioner should compute from a coarse map, how fine a map must be, and whether imagery, a different product or onboard sensing changes the answer. This paper answers these questions with a benchmark on six North American dune fields where one metre airborne lidar provides the truth.

The contributions are: (1) a benchmark that scores hazard layers against a lidar face map whose meaning is the same at every resolution, on surfaces from 2 to 60 m, on three real elevation products and with Sentinel-2 and NAIP imagery; (2) a comparison of ten families of hazard layers, every learned layer trained leave one site out; (3) a planning evaluation at operating points chosen on the other sites, with degraded face maps and a closed loop with onboard sensing; and (4) a scaling result, detection skill against face length over grid spacing.

II. Related Work

Overhead priors. Airborne lidar and overhead imagery at metre scale have served as route priors for ground vehicles [8], [9]. Drones can map ahead of the robot [10] or learn terrain properties from the air [11]. Planetary missions plan from orbital data [12], [13]. These priors are at metre scale; we ask what a coarse prior, the only free kind for most deserts, hides.

Slip on sand. Geometric costmaps score slope and roughness on a grid [14], [15], [16], [17]. On loose sand slip rises steeply with slope [5], [18]; Curiosity met high slip crossing megaripples [6]; robophysical studies relate slope and granular flow [19]. Slip can be learned from appearance [20], desert vehicles have planned on overhead and onboard maps [21], and planners propagate slip uncertainty into routes [22], [23], [24], but they assume that the map resolves the slope, which fails on dunes at 30 m.

Dunes in elevation models. Coarser grids understate slopes [7]. Dune morphometry from DEMs and imagery is well developed [25], and deep learning maps crestlines from DEM derivatives [26]. Global 30 m products have been ranked against lidar [27]; the radar mission TanDEM-X, the source of the Copernicus GLO-30 DEM, is weaker over sand [28], and open DEM error in a dune sea grows above 15 degrees [29]. Learned super resolution can sharpen coarse DEMs [30], [31]. We know of no benchmark that scores coarse map hazard

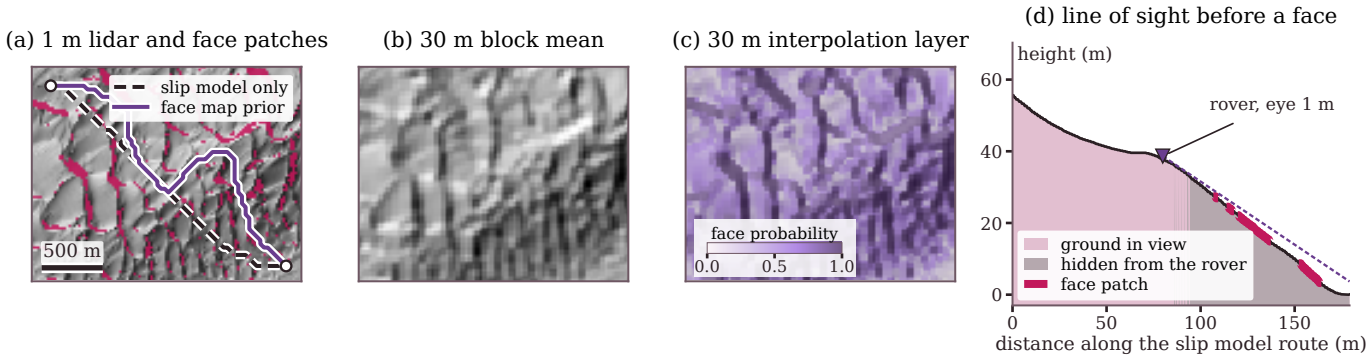


Fig. 1. What a 30 m map hides, at Great Sand Dunes. (a) One metre lidar hillshade with face patches (rose) and the 30 m routes of one E1 pair: the slip model alone (dashed) crosses faces that a face map prior (solid) avoids. (b) The 30 m block mean of the same ground, the map the planner sees. (c) The calibrated interpolation layer computed from it. North is up. (d) Lidar profile along the slip model route where it descends across faces (vertical scale exaggerated): from an eye 1 m above the ground on the slope above, the face patches lie in ground hidden behind the convex brink.

TABLE I
LIDAR WINDOWS (USGS 3DEP ONE METRE DEM; FLOWN: ACQUISITION PERIOD OF THE LIDAR PROJECT), THE NUMBER OF FACE PATCHES AND THE SHARE OF CELLS HOLDING A FACE PATCH POST AT 10 AND 30 M.

Site	Lat., lon.	km^2	Relief	Flown	Patches	10 m	30 m
Dumont Dunes, CA	35.65, -116.27	14.4	322 m	2024 to 25	4693	36 %	49 %
White Sands, NM	32.82, -106.29	6.3	20 m	2020	789	15 %	40 %
Kelso Dunes, CA	34.91, -115.72	9.7	240 m	2024 to 25	2184	15 %	38 %
Great Sand Dunes, CO	37.75, -105.54	14.8	246 m	2019 to 20	2065	16 %	32 %
Algodones Dunes, CA	32.92, -115.10	20.0	122 m	2021 to 23	3131	10 %	27 %
Little Sahara, UT	39.70, -112.39	6.4	99 m	2020 to 21	1776	11 %	35 %

layers and planners for slip faces against lidar truth across resolutions.

III. BENCHMARK

A. Sites, truth and label

Six windows of one metre bare earth lidar from the 3D Elevation Program (3DEP) of the U.S. Geological Survey (USGS) [32] cover active dune fields (Table I), chosen from 20 m slope images. A post is *steep* when its Horn slope [33] at 1 m is 28° or more, a little below the angle of repose of dry sand [4]; voids never count as steep. A *face patch* is an 8 connected group of at least 10 steep posts (10 m^2). Face patches are steep ground, not verified slip faces. For a coarse cell c with footprint F_c the label is the face map

$$y_c = \begin{cases} 1 & \text{if some } x \in F_c \text{ lies in a face patch,} \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

the label a planner needs; its meaning does not change with the cell size, although its prevalence does (Table I). An older cell label, at least 10 steep posts in F_c, means 10 % of a 10 m cell but 1.1 % of a 30 m cell and is reported only for comparison.

B. Inputs

Idealised: block means of the lidar at 2, 5, 10, 15, 20, 30 and 60 m (I10 and I30 in the tables) (no product noise, time gap or misregistration). *Real:* the USGS 1/3 arc second DEM (R10, about 8×10 m) [34], NASADEM (N30, February 2000) [35] and Copernicus GLO-30 (G30, 2011 to 2015) [1], each scored on its own geographic grid, with labels from the lidar posts inside each native cell; no horizontal shift is applied. For planning, GLO-30 is interpolated to the 30 m lidar grid. The source footprints that USGS publishes with each tile show that R10 was made from the same lidar collection as the truth at all six sites, so R10 tests the product chain, not independent data. *Imagery:* one Sentinel-2 level 2A (surface reflectance) scene per site [36] (10 m; lidar year, cloud below 5 %, lowest sun, chosen by rule) and National Agriculture Imagery Program (NAIP) images [37] (0.3 to 0.6 m, resampled to 1 m).

C. Hazard layers

Table II lists the layers, among them a logistic subgrid detector trained only on synthetic dune profiles, which ships with the planner and is one baseline among others. Every layer trained on lidar is trained leave one site out [38]; the detector is trained on synthetic profiles only.

D. Planner

The planner searches the 8 connected grid of the input. For an edge with along track grade $\tan\alpha$ (uphill positive) and cross slope $\tan\beta$,

$$d = \sqrt{(\tan\alpha + f_r)^2 + \tan^2\beta}, \qquad i = -i_0 \ln\left(1 - d/\tan\phi\right) \tag{2}$$

give the traction demand and the slip ($i = \infty$ when $d \geq \tan\phi$), with friction angle ϕ, motion resistance coefficient f_r and slip scale i_0; the slip law is the exponential shear relation of Janosi and Hanamoto [39], [40] written for the ratio of demand to capacity. Edge costs use the nominal parameters: an edge is passable when $i \leq i_{\max}$ and costs $\ell/(1-i)$ for length ℓ. For the classes, 128 parameter sets are drawn; a cell is passable for a draw when at least four of its eight headings are, q_c is the share of draws for which it is, and a cell is BLOCKED when $q_c \leq 0.05$, TRAVERSABLE when $q_c \geq 0.95$ and its face probability $p_c < \tau$, and UNKNOWN otherwise. The main vehicle, the *posterior vehicle*, takes ϕ, f_r and i_0 from a posterior of the exponential family fitted by Markov chain

Monte Carlo to twelve published slip curves of other wheeled vehicles (Mars Science Laboratory test rovers in Mars Yard sand and at Dumont Dunes [5], Scarab and El Dorado on tilted sand [41], Spirit on Husband Hill [42]; 5 to 95 % ranges ϕ 30.6 to 43.7°, f_r 0.03 to 0.32, i_0 0.33 to 1.21) and $i_{\max}$ from an assumed range of 0.4 to 0.7 (nominal values 36.9°, 0.13, 0.63 and 0.5; nominal climbing limit 15.9°). A vehicle with narrower assumed ranges (limit 25.5°) is the sensitivity. Three policies share the graph: *slip only*, *gate* (TRAVERSABLE cells only) and *weight* (no BLOCKED cell, cost times $1 + w\,p_c$ of the cell entered); gate and weight refuse descents of 28° or more. Every layer is run at $\tau \in \{0.02, \ldots, 0.9\}$ and $w \in \{1, \ldots, 100\}$ and reported at the setting that maximises mission success on the other five sites.

E. Scoring and closed loop

One hundred pairs per site, at least 1 km apart, are drawn twice: E1 from cells whose lidar posts are all gentle (truth used) and E2 from cells whose coarse slope is below 15° (no truth). A route, the polyline through its cell centres sampled every 0.5 m on the lidar, is *unsafe* when a sample falls on a face patch, and a *mission success* when it exists and is not unsafe. In the closed loop the rover follows the plan on a 5 m grid that inherits the coarse classes and face probabilities ($w = 5$), with slip costs from the coarse surface interpolated to 5 m; at every step it senses the face patches within a radius R from the lidar, blocks those cells, drops the face penalty of sensed free cells and replans when its route is blocked, within a window 300 m around the pair and up to 300 replans.

F. Statistics and prediction log

Detection is scored by the area under the receiver operating characteristic curve (AUC) [43]; AUC intervals come from a spatial block bootstrap [44] with 300 m blocks; route outcomes carry two stage bootstrap intervals (sites, then pairs). With six sites, an exact sign flip test over sites can reach at best $p = 1/32$, and $k/32$ after Holm correction [45] over k comparisons, so no site level comparison here can be significant at 0.05; we report per site counts and intervals instead and make no site level significance claim. On one core of a cloud x86 machine, assessing a 10 km^2 window and planning 100 routes on it take 0.2 and 0.2 s at 30 m. The design of every result in Tables III and IV was committed to a prediction log before it ran, with numeric predictions for 43 of them, but after earlier results on the same sites were known; the checks in Section IV-E are post hoc. The predictions of stages 25 to 28 (31 hits, 12 misses) and three amendments made before the affected results were known are in the supplement.

IV. Results

A. Which layer sees faces

Table III gives the mean AUC of every layer for the face map label. On 10 m block means most layers rank face cells well (interpolation 0.97, boosting 0.97, the detector 0.95, slope 0.90), and the USGS 1/3 arc second product, which USGS made from the same lidar, behaves almost like the 10 m block mean (interpolation 0.96). At 30 m every elevation layer loses skill. With the default settings of the learned layers, interpolation is best (0.82), ahead of boosting (0.81) and the network (0.80); the detector (0.74) is above only slope, NAIP features alone and the aspect rule, and level with Sentinel-2 features alone. Other settings help the learned layers: learned super resolution of the 30 m surface reaches 0.84 and the network trained four times longer 0.84, the best layers from elevation alone at 30 m, and boosting with five more features, among them the interpolated slope itself, 0.83. Nested tuning of boosting changes nothing (0.81). The information is still in the 10 m data: the largest of the nine 10 m interpolation scores in a 30 m cell reaches 0.96.

TABLE II
HAZARD LAYERS. L: TRAINED ON THE LIDAR OF THE OTHER FIVE SITES (LEAVE ONE SITE OUT), ON ITS FACE LABELS OR, FOR SUPER RESOLUTION, ON ITS ELEVATION. EVERY LAYER THAT IS NOT A PROBABILITY IS CALIBRATED BY LOGISTIC REGRESSION [46] ON THE OTHER FIVE SITES WHEN A PLANNER NEEDS ONE.

Layer	Score of a cell	L
Slope	Horn slope of the coarse surface, the input of a slope costmap	
Ruggedness	Terrain ruggedness index [47] over the 8 neighbours	
Roughness	Root mean square residual of a plane fitted to 3×3 cells, as in GESTALT [16]	
Aspect rule	Slope times agreement of the cell's downslope azimuth with the dominant face direction of the site (from the coarse map)	
Interpolation	Steepest slope after cubic spline interpolation of the coarse surface to 1 m	
Super resolution	Steepest slope of a 5 m surface predicted by a dilated network [48] as a residual [49] of the interpolated 30 m surface	yes
Detector	Logistic model on slope, convexity, slope break and relief, trained on 60 000 synthetic straight crested dune profiles sampled at the cell size	
Boosting	Gradient boosted trees [50], [51], [52] on 14 terrain features (slope, ruggedness, roughness, curvature, topographic position, relief, detector features); variants with 5 more features (Boosting+), tuned by nested leave one site out, and without the detector features	yes
Network	Fully convolutional network [53], four 3×3 layers dilated by 1, 2, 4 and 8 [48], trained with Adam [54] for 1500 or 6000 steps	yes
+ imagery	Boosting on the terrain features plus Sentinel-2 or NAIP features (band means, brightness, contrast, shadow share); also imagery alone	yes

Imagery adds information that elevation lacks. Boosting on terrain plus Sentinel-2 features reaches 0.85 at 30 m (+0.04 over terrain alone), and with NAIP 0.84; imagery alone is weaker (0.74 and 0.67), and at 10 m it adds nothing (0.97). Super resolution and boosting with Sentinel-2 are above interpolation at all six sites and the detector below it at all six, but with six sites none of the seven comparisons against interpolation that we tested at the site level (detector, boosting, network, super resolution, boosting with curvature features, with Sentinel-2 and with NAIP) is significant after Holm correction (sign flip $p = 0.031$ at best, adjusted 0.22).

The real 30 m products are worse than the 30 m block mean: on their own grids the best layer reaches 0.72 on GLO-30 and 0.63 on NASADEM, which predates the lidar by about two decades; interpolating GLO-30 to the lidar grid changes little (best 0.72). Much of this loss is dune migration between the radar and lidar acquisitions rather than product error (supplement S6).

TABLE III

DETECTION OF THE FACE MAP LABEL OF EQ. (1): MEAN AUC OVER THE SIX SITES. I: LIDAR BLOCK MEANS; I10→30: THE LARGEST OF THE NINE 10 M SCORES IN EACH 30 M CELL; R10: USGS 1/3 ARC SECOND DEM, RESAMPLED BY USGS FROM THE SAME LIDAR AT ALL SIX SITES; N30: NASADEM; G30: GLO-30; R10, N30 AND G30 ON THEIR OWN GRIDS, WHERE THE NETWORK AND SUPER RESOLUTION WERE NOT RUN (ON GLO-30 INTERPOLATED TO THE 30 M LIDAR GRID THEY REACH 0.72 AND 0.70). *TRAINED ON THE LIDAR OF THE OTHER FIVE SITES. BEST PER COLUMN IN BOLD (TIES AT TWO DECIMALS ALL BOLD); · NOT RUN.

Layer	I10	I30	I10→30	R10	N30	G30
Slope	0.90	0.69	·	0.90	0.62	0.65
Ruggedness	0.94	0.76	·	0.93	**0.63**	0.69
Roughness	0.80	0.77	·	·	·	·
Aspect rule	0.83	0.61	·	·	·	·
Interpolation	**0.97**	0.82	**0.96**	**0.96**	0.62	0.70
Super resolution*	·	0.84	·	·	·	·
Detector	0.95	0.74	0.95	0.95	0.56	0.65
Boosting*	**0.97**	0.81	**0.96**	0.95	**0.63**	**0.72**
Boosting+*	**0.97**	0.83	·	·	·	·
Boosting, tuned*	**0.97**	0.81	·	·	·	·
Network*	0.96	0.80	0.95	·	·	·
Network, 6000 steps*	·	0.84	·	·	·	·
Sentinel-2 only*	0.68	0.74	·	·	·	·
Boosting + Sentinel-2*	**0.97**	**0.85**	·	·	·	·
NAIP only*	0.62	0.67	·	·	·	·
Boosting + NAIP*	**0.97**	0.84	·	·	·	·

The label matters for small faces. With the earlier cell label (at least 10 steep posts) the 30 m interpolation scores 0.82; requiring 100 or 300 steep posts raises it to 0.89 and 0.97, so a 30 m map ranks large faces well and small ones poorly; a 30 degree threshold changes little (0.82). The labels are spatially correlated over 30 to 870 m (the largest at Dumont, whose window holds rocky hills); with 870 m blocks the per site intervals of slope and the detector still lie below interpolation at all six sites.

B. How fine a map must be

Fig. 2a shows how detection falls with grid spacing. Every layer is almost perfect at 2 m (interpolation 1.00) and still good at 10 m (0.97); between 10 and 30 m interpolation falls to 0.82 and at 60 m to 0.69. Fig. 2b relates the fall to face size. At every spacing, boosting finds nearly all faces longer than four grid cells (96 % of faces 4 to 8 cells long at 30 m, 90 % at 60 m), but at 30 and 60 m far fewer faces shorter than half a cell (60 % at 30 m, 46 % at 60 m). Longer patches touch more cells and so have more chances to be flagged, which accounts for part of this rise. Most faces in these fields are short at 30 m: 75 % of the patches are shorter than one 30 m cell.

C. Planning

Table IV gives mission success at the operating points chosen on the other five sites, and supplement S4 shows the weight policy at every weight. With the posterior vehicle the gate routes at most 34 % of pairs at 10 m and 42 % at 30 m, whatever the layer, because the slip model is uncertain on most dune slopes (most refusals are at an endpoint: 76 % for boosting at 30 m and $\tau = 0.2$). The weight policy routes almost every pair and is chosen for every layer; on 10 m maps it mostly takes the largest weight of the grid ($w = 100$).

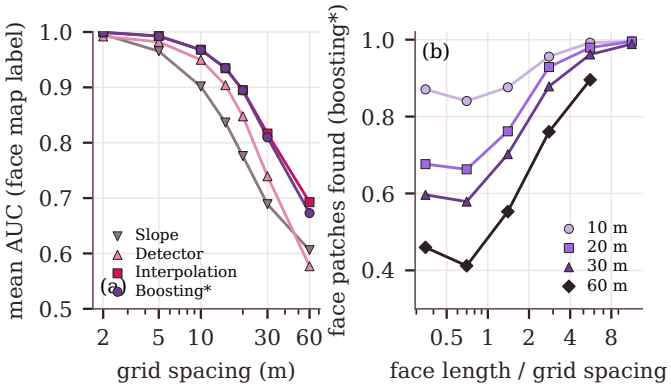


Fig. 2. Resolution and scale (lidar block means, face map label). (a) Mean AUC over six sites against the grid spacing. (b) Share of face patches found by boosting (a cell holding the patch is flagged at the threshold that flags 10 % of face free cells) against patch length over grid spacing, in bins with edges 0.5, 1, 2, 4 and 8, for four spacings; bins with fewer than 50 patches are omitted. Patch length: major axis of the ellipse with the same second moments. *Trained leave one site out.

On 10 m maps every layer helps. Mission success rises from 17 % with the slip model alone to 64 % with interpolation (+47 points, two stage interval +28 to +64, positive at all six sites), 59 % with boosting and 50 % with the detector. On 30 m maps the gain is small: interpolation adds +5 points (interval +2 to +9) and boosting with Sentinel-2 +5 (interval +1 to +9) to 8 %, and no layer computed from GLO-30 has an interval above zero. With the assumed vehicle the 30 m gain of interpolation is +7 points. Endpoints chosen without the truth (E2) lower every rate, the slip model alone to 3 % at 30 m, and there the gain of interpolation is no longer clearly above zero (interval 0 to +6).

The face map, an upper bound that assumes face information from fine data, reaches 81 % on the 30 m grid. The registered degraded face maps (Table IV, lower part, at $w = 5$; the gate routes too few pairs with this vehicle to compare) show how good that information must be: the face map gives 55 % at 30 m, displaced by 5 m 50 %, by 10 m 43 % and by 30 m 11 %. Face maps built with the same 28 degree rule from simulated 2 m and 5 m surveys give only 23 % and 8 %; supplement S6 shows why. The outcome is robust to the scoring details (supplement S5).

D. Onboard sensing

Table V adds onboard sensing that sees every face within the radius, an idealised bound. With $R = 20$ m the rover reaches the goal without touching a face on 80 % of the pairs when its 30 m prior holds only the slip model, and on 84 % with boosting as prior (+4 points, interval -1 to +13); the face map prior gives 85 %. No route between E1 endpoints touched a face and no pair used up its replan budget; every failure is a pair left without a route, 4 of 120 at 30 m and 17 at 10 m already before the first move (the same pairs for every prior, which is why the 10 m prior does no better). A better prior mainly saves replanning (29 replans per pair with the slip model prior, 8 with the face map). With E2 endpoints 5 % of

TABLE IV

MISSION SUCCESS (%) AT THE OPERATING POINT CHOSEN ON THE OTHER FIVE SITES (GATE THRESHOLD OR WEIGHT), 100 PAIRS PER SITE, POSTERIOR VEHICLE. E1: ENDPOINTS CHOSEN WITH THE TRUTH; E2: CHOSEN FROM THE COARSE MAP ONLY. †TWO STAGE BOOTSTRAP INTERVAL OF THE GAIN OVER THE SLIP MODEL ENTIRELY ABOVE ZERO (NOT CORRECTED FOR THE NUMBER OF CELLS). LOWER PART, A SECOND PROTOCOL: FACE MAPS DEGRADED AS REGISTERED, WEIGHT POLICY AT A FIXED $w = 5$. · NOT RUN. *TRAINED LEAVE ONE SITE OUT.

	I10		I30		G30
Layer	E1	E2	E1	E2	E1
Slip model only	17	9	8	3	6
Slope	35†	28†	6	3	6
Interpolation	64†	53†	13†	5	7
Detector	50†	33†	7	4	5
Detector, recalibrated	52†	37†	5	2	6
Boosting*	59†	47†	9	4	7
Network*	51†	40†	6	3	7
Boosting + Sentinel-2*	56†	46†	13†	7†	·
Face map (oracle)	84†	80†	81†	68†	80†
Face map, $w = 5$	81	71	55	40	·
Face map 5 m off	67	56	50	39	·
Face map 10 m off	36	26	43	32	·
Face map 30 m off	19	10	11	5	·
Face map from 2 m survey	29	19	23	14	·
Face map from 5 m survey	17	9	8	4	·

TABLE V

CLOSED LOOP WITH ONBOARD SENSING (RADIUS R), 20 PAIRS PER SITE, POSTERIOR VEHICLE, SENSING EVERY FACE WITHIN THE RADIUS: SUCCESS (%; GOAL REACHED, NO FACE CONTACT) AND REPLANS PER PAIR (RP), E1 ENDPOINTS UNLESS MARKED E2. MAIN RUNS $R = 20$ M; I30 ALSO WITH $R = 10$ AND 30 M.

	I10, R 20		I30, R 20		I30, S		
Prior layer	S	Rp	S	Rp	E2	R 10	R 30
Slip model only	78	11	80	29	67	76	83
Detector	81	5	85	25	73	82	85
Interpolation	80	4	82	20	71	81	87
Boosting*	80	4	84	21	71	81	88
Face map	80	1	85	8	75	83	86

the pairs touch a face whatever the prior. Real faces lie behind crests (Fig. 1d, Section IV-E).

E. Post hoc checks

These checks were decided after a second review and are post hoc. *Line of sight.* When the rover senses a 5 m cell only if the line from an eye 1 m above the ground to 0.2 m above the cell centre clears the lidar (I30, E1, $R = 20$ m, 120 pairs), success falls to 30 % with the slip model prior, and 65 % of the pairs touch a face. Boosting as prior gives 42 % (+12 points, interval +3 to +21, higher at all six sites) and the face map 74 % (+44 points, interval +25 to +63).

Finer and learned planning inputs. On 5 m block means, with weights up to $w = 1000$ (interpolation chose 1000 at every site), interpolation reaches 86 % against 34 % for the slip model alone (+52 points, interval +31 to +71) and the face map 93 %. On 30 m maps, the two layers that rank faces best add little: super resolution +5 points (interval +2 to +8) and the network trained for 6000 steps +2 (interval 0 to +5).

Displaced face maps at the chosen weight. With the weight chosen on the other five sites, the face map gives 84 % on the 30 m grid, displaced by 5 m 75 %, by 10 m 61 % and by 30 m 11 %.

Larger faces. Scored only on faces of at least 300 posts, 30 m interpolation still adds only 13 points over the slip model, against 54 on 10 m maps. Supplement S6 gives the details, why the survey face maps fail, and the effect of dune migration on GLO-30.

V. DISCUSSION AND LIMITATIONS

What the results mean. A 30 m elevation model keeps enough of the dune shape to rank face cells better than chance, but not enough to plan around faces: the best 30 m layers add a few points of mission success, while the same layers on a 10 m map add up to +47 points. Fig. 2b suggests why: faces several grid cells long are found at every spacing, and most faces in these fields are shorter than one 30 m cell. Better learning helps detection a little (super resolution 0.84 against 0.82 for interpolation) and planning about as much as interpolation (Section IV-E). Imagery gave the largest gain in 30 m detection, as in dune mapping [25], but not in planning. Faces hide behind crests, so with line of sight a good prior more than doubled success. Face information from fine data, placed within a few metres, would make most 30 m routes safe, so a drone survey must be well georeferenced and its face threshold calibrated for its resolution.

Limitations. The truth is steep ground: rock, vehicle ruts and vegetated mounds also exceed 28 degrees, Dumont, Algodones and Little Sahara are off road vehicle areas, and the manual audit of a stratified sample of 600 patches is not finished, so label precision is unknown. We chose the windows, used them in earlier tests of the detector, and keep the prediction log in our own repository. The vehicle model is a posterior fitted to published slip curves of other vehicles; in our own leave one study out validation no slip model family transferred between published studies (median held out slip error 0.20 to 0.41 depending on the family), so absolute route outcomes are uncertain. No rover was driven, no physics was simulated, and even the line of sight check assumes that every visible face is detected. Six sites limit the training data of the learned layers and the power of site level tests. We did not test Arabian fields: a megadune face 100 m high spans about five 30 m cells and should be visible, but smaller flank faces would not be, GLO-30 is weaker over sand [28] and dunes migrate.

VI. CONCLUSION

A 30 m elevation model ranks slip faces better than chance but cannot keep a small rover off them. On six lidar dune fields no hazard layer we planned with on such a map, including imagery fusion and learned super resolution, added more than a few points of mission success, while the same kind of layer on a 10 m map raised it from 17 % to 64 %, and the gap remained when only large faces were scored. Because faces hide behind crests, onboard sensing does not remove the need for a good prior: with line of sight a face map prior raised success from 30 % to 74 %. Next are the label audit, Arabian

fields with a drone survey, and a rover with measured slip on the same sand.

Data and Code

USGS 3DEP lidar and DEMs, NASADEM and NAIP are in the public domain; Sentinel-2 data are provided under the Copernicus free licence; Copernicus GLO-30 is used under its licence (produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA; all rights reserved). The code, result files, prediction log and the scripts that regenerate every number and figure of this paper and its supplement are at https://github.com/hudasol/dune-rover (stages 25 to 29).

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